

I-STEM: Strengthen the Indigenous Development of Scientific Instruments

After establishing a National web portal through which R&D equipment in institutions across the country can be located and accessed in a transparent way, I-STEM is entering a New Phase with the following objectives –

- ▶ Working towards a more comprehensive listing of R&D facilities. Availability, and their active use
- ▶ Providing partial assistance to Tier 2 and Tier 3 institutions for the maintenance of facilities shared through I-STEM through Comprehensive AMC program
- ▶ Partial assistance to Users from Tier 2 and Tier 3 institutions who use facilities through I-STEM
- ▶ Promoting the indigenous development – especially by startups – of scientific equipment and value-added technical supplies, including software, used for R&D; enabling such development via the (national) Digital Catalogue of indigenously developed technologies and products maintained by the I-STEM portal
- ▶ Establishing regional centres for providing technical training and skill development, in part by restoring/stripping R&D equipment written off by I-STEM institutions; thus developing a stream of talent and gainful employment through maintaining and upgrading R&D facilities
- ▶ Providing access through the web to R&D software platforms to Academic Users to promote R&D and more effective instruction/learning, especially in smaller institutions.



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tion remains a major issue. Though an improvement over ultrasonic sensors, the range and resolution of radars still remain an issue.

3. Lidar pricing, range, and issues with bad weather. The sensors that the autonomous vehicles most heavily depend upon are lidars. With their higher resolution and accuracy, lidars are, in a sense, what made self-driving vehicles a reality. But it is this higher accuracy of lidars that has become its own enemy.

Lidars are not good at handling rain. As we have already discussed, lidars overcome the pitfalls of radars by firing faster laser pulses that then get reflected back to the sensor when they hit an obstacle. This gives lidar the information of the object, its shape, size, and distance from the vehicle. Turns out, lidars are fast enough that their laser pulses hit the droplets of water falling in the rain. When these laser pulses are reflected back to the lidar, it mistakes the droplets of water for an object. Correcting this critical error is still an open prob-

lem and expensive both in terms of computational power and resources.

Additionally, lidars have the same range as radars, that is, 200 metres. Though more efficient than radars, lidars can look no further than radars in terms of distance. Another complication is the pricing. Lidars can be extremely expensive, thus making them harder for companies to install as they increase the overall pricing of the product.

4. Range, resolution, and complexity of cameras. High-resolution cameras are still an economical option as compared to lidars, but they do come with a set of unique challenges of their own. For starters, there is a huge gap between the ratio of range and resolution between ultrasonic sensors, radars, lidars, and high-definition cameras. Needless to say, at a longer distance, cameras can give you a higher resolution as compared to any other sensor used. With each sensor giving its own limited resolution given its capacity and the added complexity of the high resolution of the camera,

defining an object is an extremely computationally expensive task.

Additionally, with the given time limit in decision making (which can be milliseconds) and the lack of infrastructure, especially of the 5G Internet, this problem is further compounded. Another major problem with images obtained from cameras is that they require very advanced machine learning algorithms to run with the help of huge datasets. This creates a two-fold problem. First is the allocation of the computational resources and second is the allocation of memory and storage.

Human/societal limitations

As mentioned earlier, we are not ready for the autonomous cars yet. Following are some of the hurdles we face:

1. We don't have the mindset for self-driving cars. For over a hundred years, cars have been created to be driven. As such, if you know how to drive, then purchasing an autonomous car requires a complete shift in mindset from you. You know how you hit